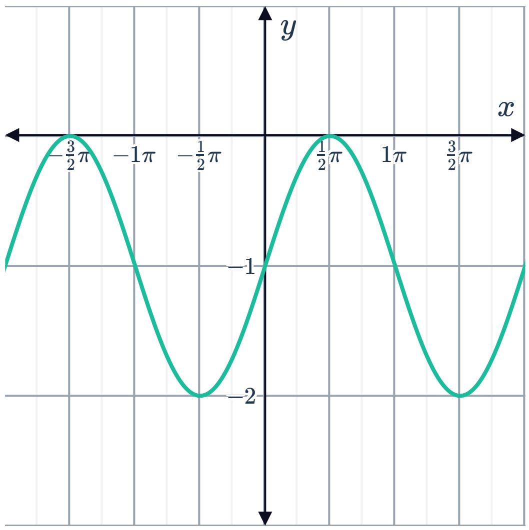
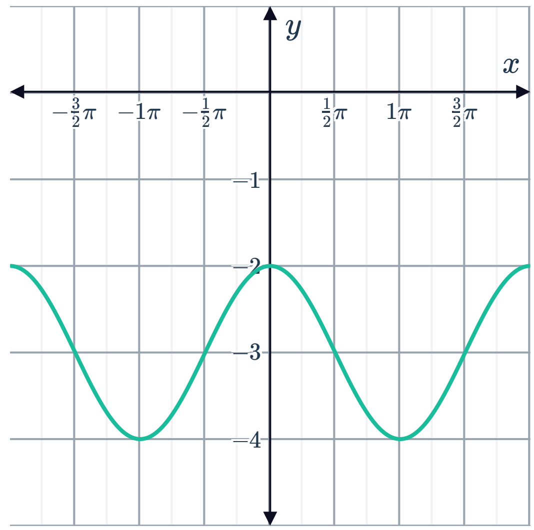
Translations of sine and cosine functions

1 Find the equation of each of the graphed functions:

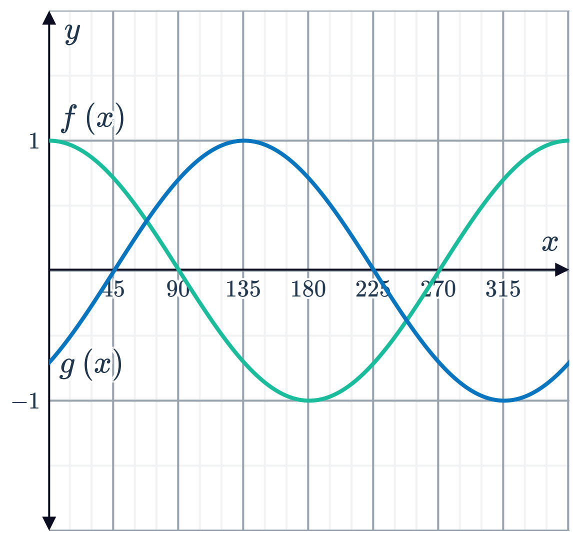
a



b



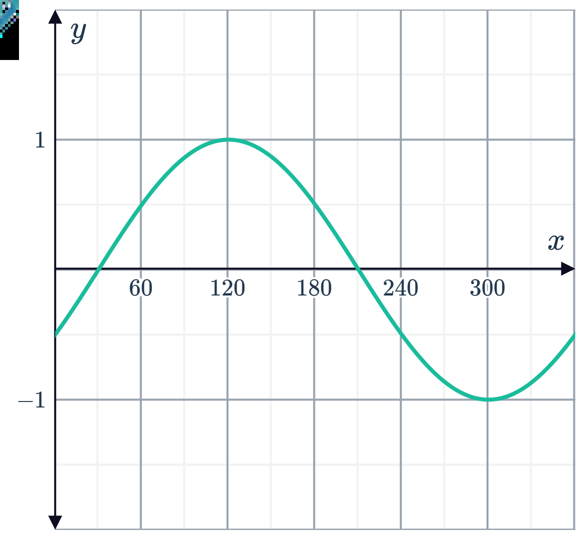
2 The functions $f\left(x\right)$ and $g\left(x\right)=f\left(x+k\right)$ have been graphed on the following coordinate plane:



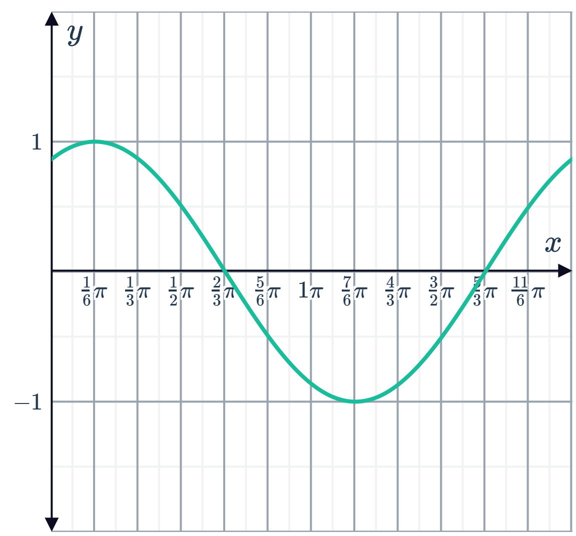
a Describe the transformation occurred from $f\left(x\right)$ to $g\left(x\right)$.

b Determine the smallest positive value of k.

3 Find the equation of the graphed function given that it is of the form $y = \sin(x - c)$, where c is the least positive value and x is in degrees.



4 Find the equation of the graphed function given that it is of the form $y = \cos(x - c)$, where c is the least positive value.



5 The graph of $y = \sin x$ is translated 60° to the left.

a Find the equation of the new curve.

b State the amplitude of the new curve.

c State the period of the new curve.

6 The graph of $y = \cos x$ is translated $\dfrac{\pi}{3}$ units to the left.

a Find the equation of the new curve.

b State the amplitude of the new curve.

c State the period of the new curve.

7 Determine the values of c in the region $-360^\circ \leq c \leq 360^\circ$ that make $y = \sin(x - c)$ the same as $y = \cos x$.

8 Consider the function $y = \sin x - 2$.

a Determine the period of the function, in degrees.

b Determine the amplitude of the function.

c Determine the maximum value of the function.

d Determine the minimum value of the function.

e Sketch the graph of the function.

9 Consider the function $f(x) = \sin x$ and $g(x) = \sin(x - 90^\circ)$.

a Complete the table of values:

x	0	90°	180°	270°	360°
$f(x)$					
$g(x)$					

b Describe the transformation of the graph of $f(x)$ that results in the graph of $g(x)$.

c Sketch the graphs of $f(x)$ and $g(x)$ on the same coordinate plane.

10 Consider the function $f(x) = \cos x$ and $g(x) = \cos\left(x - \frac{\pi}{2}\right)$.

a Complete the table of values:

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$					
$g(x)$					

b Describe the transformation of the graph of $f(x)$ that results in the graph of $g(x)$.

c Sketch the graphs of $f(x)$ and $g(x)$ on the same coordinate plane.

11 Consider the functions $y = \cos x$ and $y = \text{co}$[redacted]$\frac{\pi}{4}\Big)$.

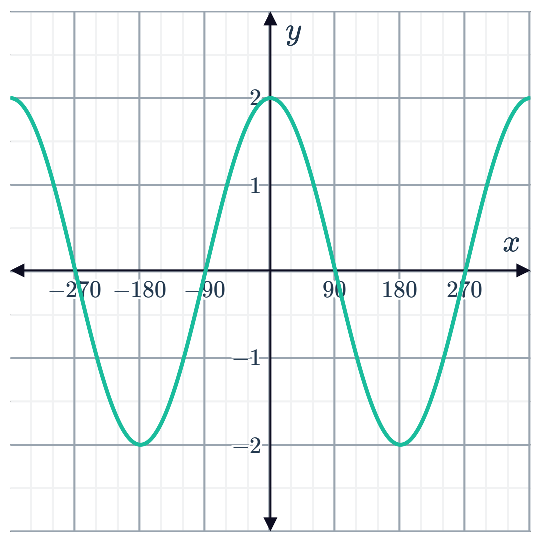
a Complete the following table of values, rounding your answers to two decimal places where necessary:

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$								
$\cos\left(x + \frac{\pi}{4}\right)$								

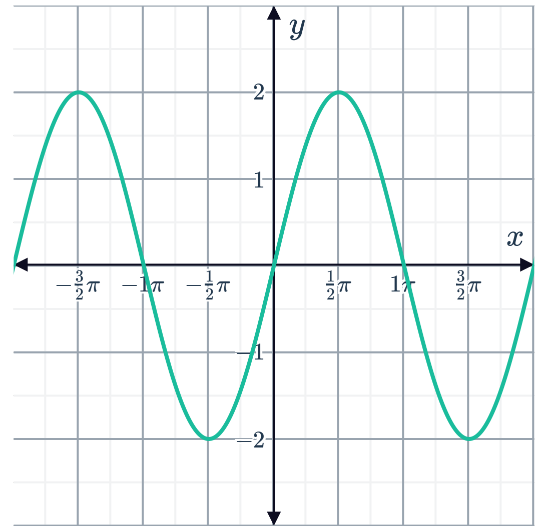
b Are the values of $\cos\left(x + \frac{\pi}{4}\right)$ values of $\cos x$ shifted $\frac{\pi}{4}$ to the left or to the right?

Dilations of sine and cosine functions

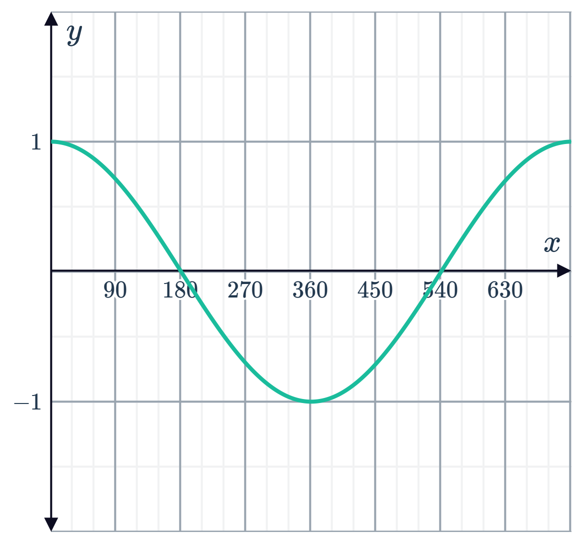
12 Find the equation of the graphed function:



13 Find the equation of the graphed function:



14 Find the equation of the graphed function given that it is of the form $y = \sin bx$ or $y = \cos bx$, where b is positive, and x is in degrees.



15 Determine whether the following transformations represent a change in the period of the function $y = \sin x$:

a $y = \sin\left(\frac{1}{2}x\right)$ **b** $y = \sin(2x)$ **c** $y = \sin x + 2$ **d** $y = 2\sin x$

e $y = \sin(x - 2^\circ)$ **f** $y = \sin(5x)$ **g** $y = \sin(x - 5)$ **h** $y = 5\sin x$

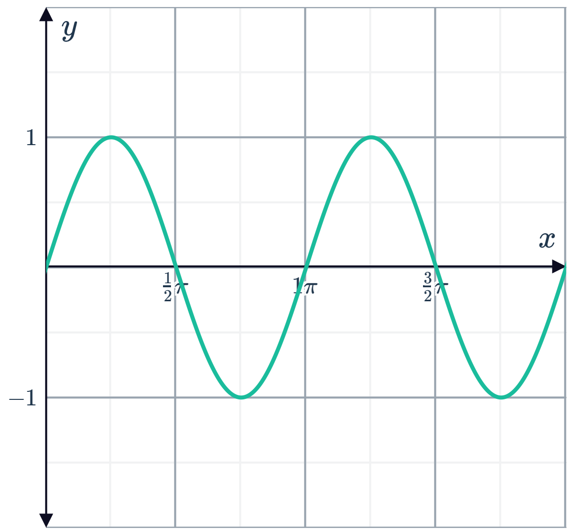
i $y = \sin\left(\frac{1}{5}x\right)$ **j** $y = \sin x + 5$

16 Consider a sine function y with an amplitude of 3 and a period of $\frac{\pi}{3}$.

a Find the equation for this function. **b** Sketch the graph of the function.

17 Find the equation of each graphed function[illegible]n that it is of the form $y = \sin bx$ or $y = \cos bx$, where b is positive:

a



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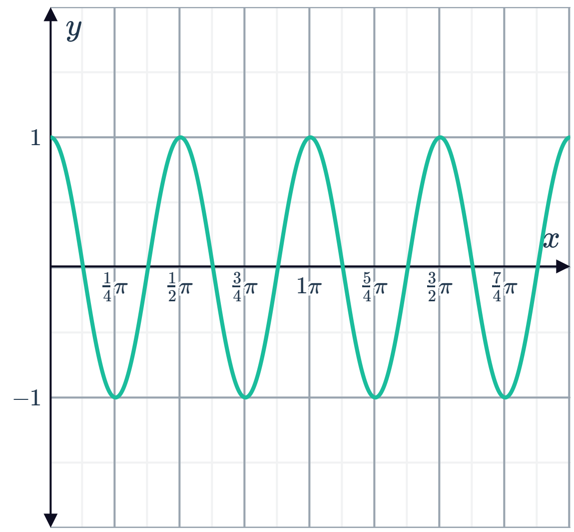


18 State the amplitude of the following functions:

a $y = -4\sin x$

b $f\left(t\right) = -\dfrac{1}{9}\sin t$

19 Determine whether each of the following transformations of $y = \cos x$ has a different amplitude to $y = \cos x$:

a $y = \cos\left(x - 9\right)$

b $y = 9\cos x$

c $y = \cos 9x$

d $y = \cos x + 9$

20 Consider the function $y = -5\cos x$.

a Find the maximum value of the function.

b Find the minimum value of the function.

c Find the amplitude of the function.

21 Consider the function $f(x) = \cos x$ and $g(x) = \cos\left(\frac{x}{2}\right)$.

a State the period of $f(x)$, in degrees.

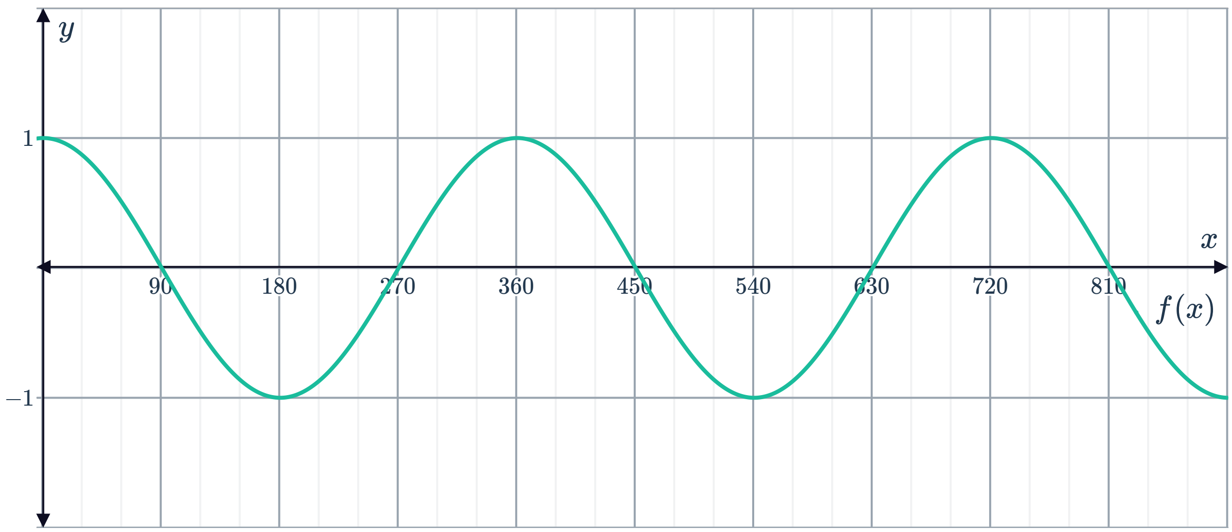
b Complete the table of values for $g(x)$:

x	0°	180°	360°	540°	720°	900°	1080°
$g(x)$							

c State the period of $g(x)$, in degrees.

d What transformation of the graph of $f(x)$ results in the graph of $g(x)$?

e The graph of $f(x)$ has been provided below. Sketch the graph of $g(x)$.

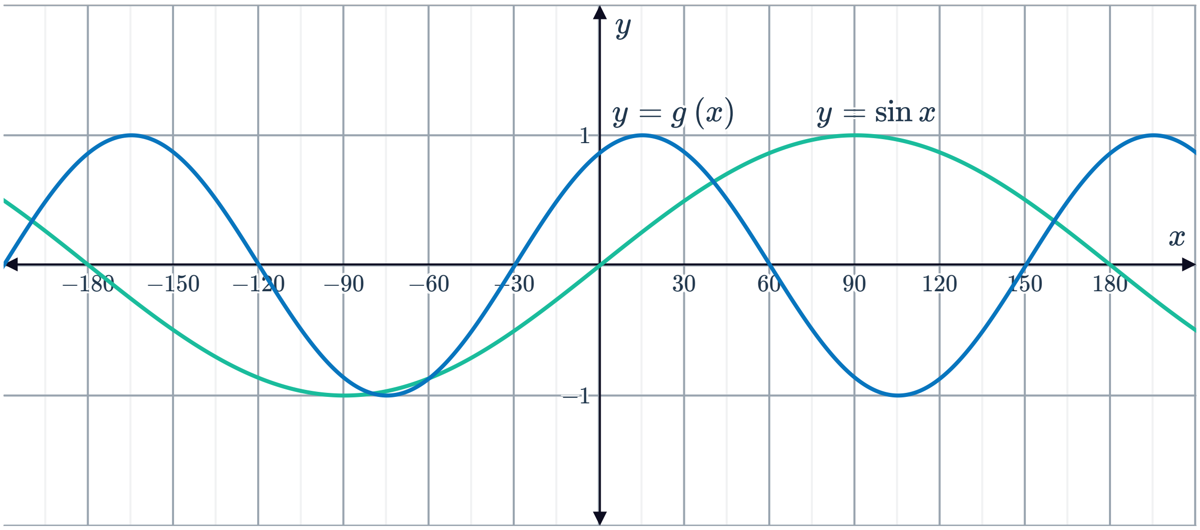


f Is the amplitude of $g(x)$ different to the amplitude of $f(x)$?

Series of transformations

22 Describe the series of transformations of the graph of $y = \cos x$ that results in the graph of $y = \cos(x + 45^\circ) - 2$.

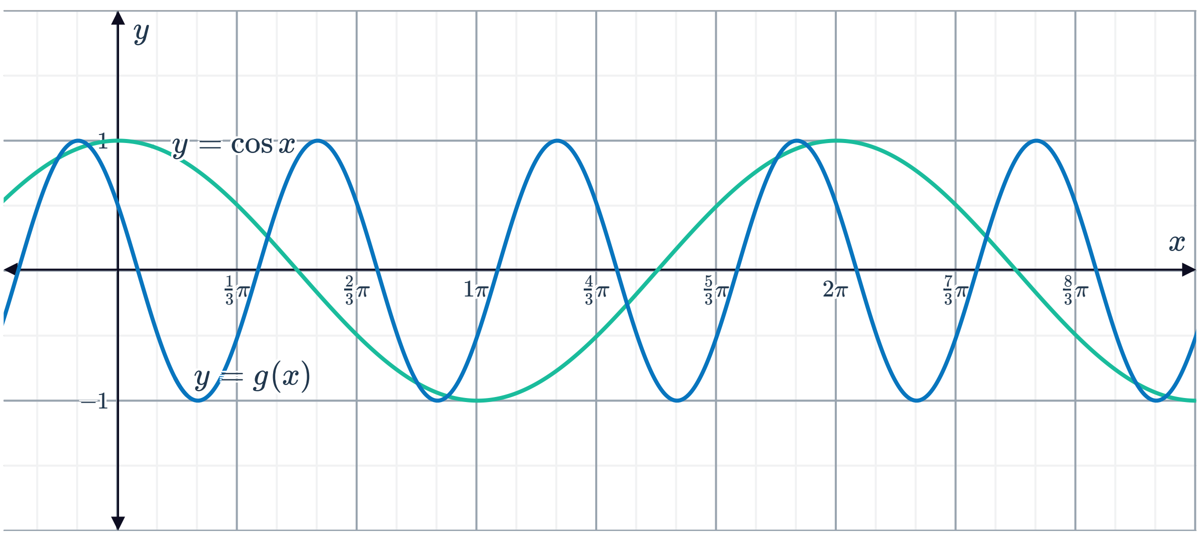
23 Consider the graphs of $y = \sin x$ and $y = g\,($[redacted]phed on the same coordinate plane:



a Describe the series of transformations required to obtain the graph of $y = g\left(x\right)$ from the graph of $y = \sin x$.

b Now, state the equation of $g\left(x\right)$.

24 Consider the graphs of $y = \cos x$ and $y = g\left(x\right)$ graphed on the same coordinate plane:



a Describe the series of transformations required to obtain the graph of $y = g\left(x\right)$ from the graph of $y = \cos x$.

b Now, state the equation of $g\left(x\right)$.

25 For each of the following functions:

i State the amplitude of the function.

ii Determine the phase shift of the function. Use a positive value to represent a shift to the right, and a negative value to represent a shift to the left.

iii Sketch the graph of the function.

a $y = 4\sin(x - \pi)$

b $y = \sin\left(x - \frac{\pi}{2}\right)$

c $y = 5\cos\left(x - \frac{\pi}{3}\right)$

d $y = -4\cos\left(x + \frac{\pi}{2}\right)$

26 Consider the graph of $y = \sin x$.

a Describe how to obtain the graph of $y = \sin(x - 90^\circ) - 2$ from the graph of $y = \sin x$.

b Sketch the graph of $y = \sin(x - 90^\circ) - 2$ and $y = \sin x$ on the same coordinate plane.

c State the period of the curve $y = \sin(x - 90^\circ) - 2$, in degrees.

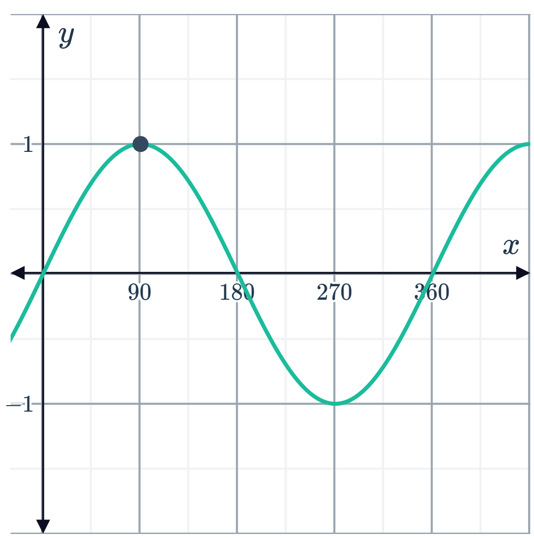
27 Consider the graph of $y = \sin x$.

a Describe how to obtain the graph of $y = \sin\left(x - \frac{\pi}{2}\right) + 3$ from the graph of $y = \sin x$.

b Sketch the graph of $y = \sin\left(x - \frac{\pi}{2}\right) + 3$ and $y = \sin x$ on the same coordinate plane.

c State the period of the curve $y = \sin\left(x - \frac{\pi}{2}\right) + 3$.

28 Consider the graph of $y = \sin x$ and its first maximum point for $x \geq 0$ at $(0, 1)$. By considering the transformation that has taken place, state the coordinates of the first maximum point of each of the given functions for $x \geq 0$:



a $y = 5\sin x$

b $y = -5\sin x$

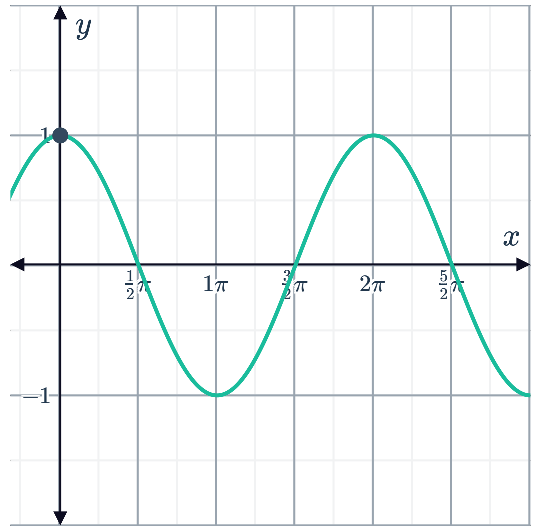
c $y = \sin x + 1$

d $y = \sin 2x$

e $y = \sin(x - 60^\circ)$

f $y = 5\sin x + 1$

29 Consider the graph of $y = \cos x$ and its first [redacted]um point for $x \geq 0$ at $(0, 1)$. By considering the transformation that has taken place, state the coordinates of the first maximum point of each of the given functions for $x \geq 0$:



a $y = \cos\left(x + \dfrac{\pi}{3}\right)$

b $y = 5\cos\left(x - \dfrac{\pi}{3}\right)$

c $y = 2 - 5\cos x$

d $y = \cos\left(\dfrac{x}{4}\right)$

e $y = 5\cos 4x - 2$

f $y = \cos\left(x - \dfrac{\pi}{3}\right) + 2$

30 For each of the following functions:

i State the domain of the function.

ii State the range of the function.

iii Sketch the graph of the function for $-180^\circ \leq x \leq 180^\circ$.

a $y = \sin 2x - 2$

b $y = -5\sin 2x$

c $y = \sin\left(\dfrac{x}{3}\right) + 5$

d $y = \cos\left(\dfrac{x}{2}\right) - 3$

31 For each of the following functions:

i State the amplitude.

ii Find the period, in degrees.

iii Sketch the graph of the function for $0^\circ \leq x \leq 360^\circ$.

a $y = \cos 3x$

b $y = \sin\left(\dfrac{x}{2}\right)$

c $y = \sin\left(\dfrac{3x}{4}\right)$

d $y = \cos\left(\dfrac{x}{3}\right)$

e $y = -\cos 4x$

f $y = 3\sin(0.25x)$

g $y = 2\sin 3x$

h $y = -5\sin 3x$

32 For each of the following functions:

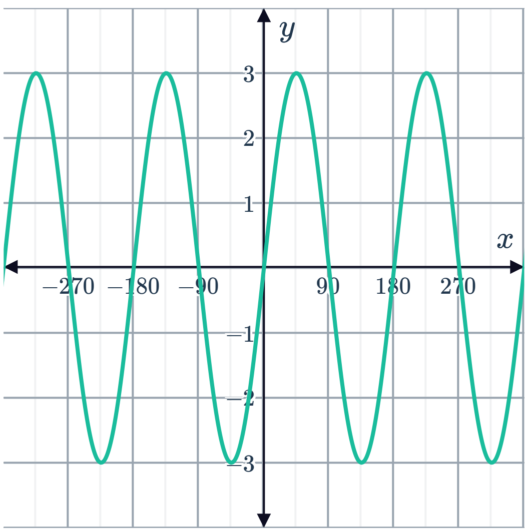
- **i** Find the period, in degrees.
- **ii** State the amplitude of the function.
- **iii** Find the maximum value of the function.
- **iv** Find the minimum value of the function.
- **v** Sketch the graph of the function on the domain $-360\degree \leq x \leq 360\degree$.

a $y = 3\sin x - 3$

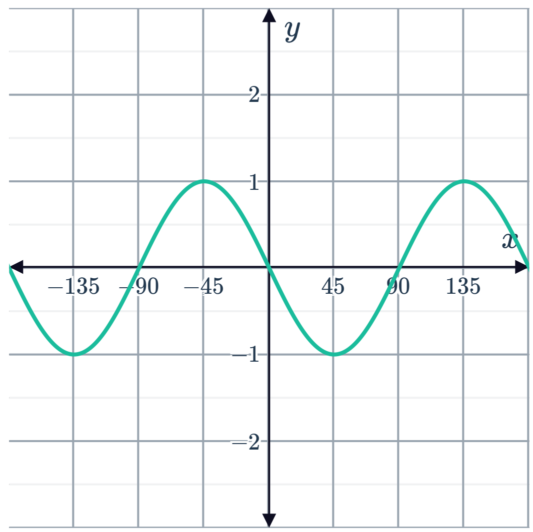
b $y = 3\sin 2x - 2$

33 For each of the following graphs, find the equation of the function given that it is of the form $y = a\sin bx$ or $y = a\cos bx$, where b is positive:

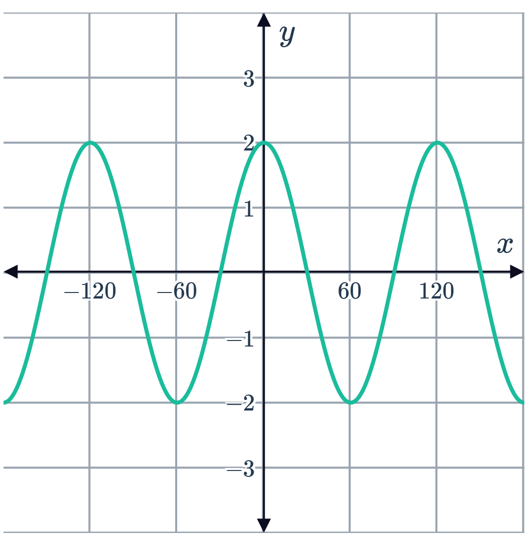
a



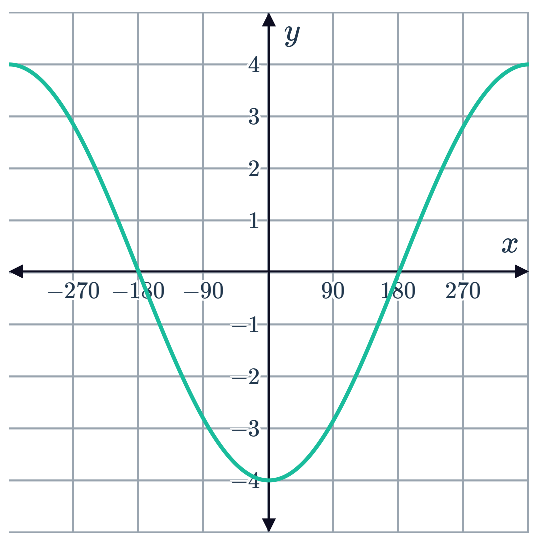
b



c



d

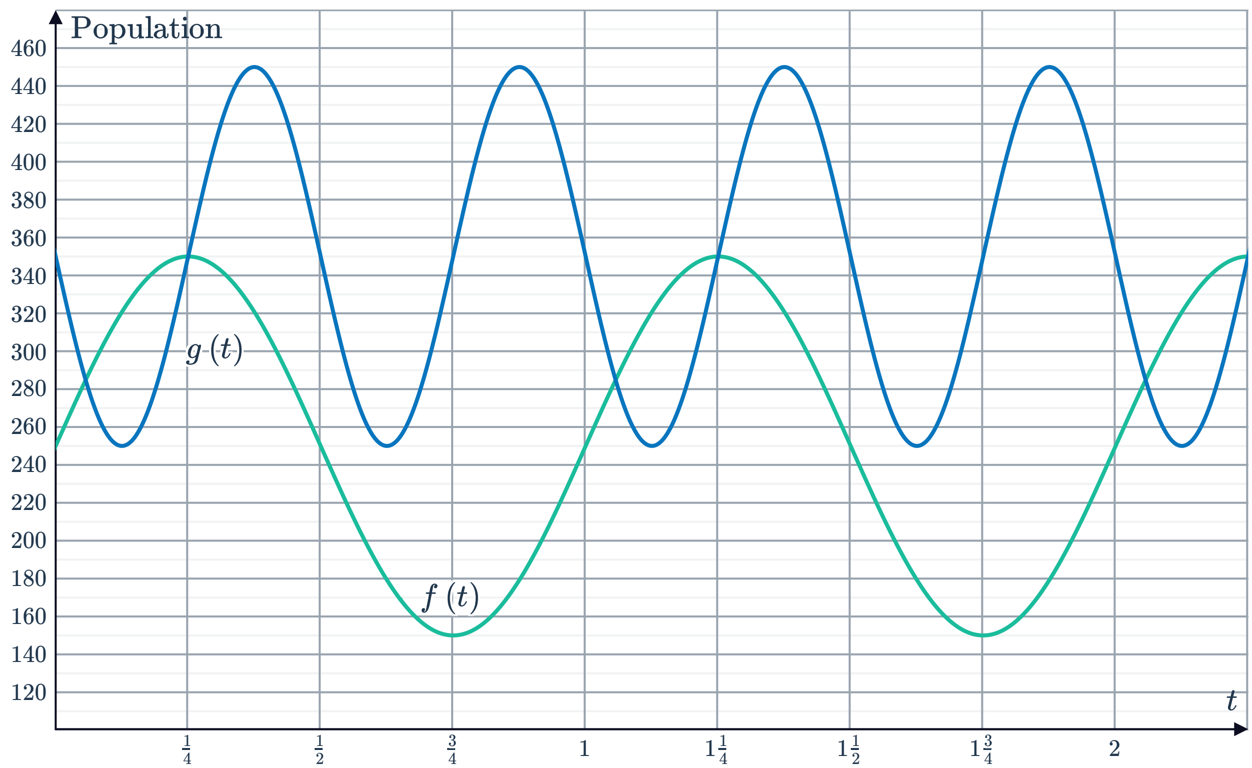


Applications

34 The population (in thousands) of two different types of insects on an island can be modeled by the following functions:

- Butterflies: $f(t) = a + b\sin(mt)$
- Crickets: $g(t) = c - d\sin(kt)$

where t is the number of years from when the populations were first measured, and $a, b, c, d, m,$ and k are positive constants. The graphs of f and g for the first two years are shown below:



a Find the function $f(t)$ that models the population of butterflies over t years.

b Find the function $g(t)$ that models the population of crickets over t years.

c Find the number of times over an 18-year period that the population of crickets reaches its maximum.

d How many years after the population of crickets first starts to increase, does it reach the same population as the butterflies?

e Find t, when the population of butterflies first reaches $200\,000$.

35 The tide rises and falls in a periodic manner [illegible]n can be modeled by a trigonometric function. Brad charts tide levels in order to determine when he can sail his ship into a bay and takes the following measurements:

- Low tide occured at 8 am, when the bay was 7 m deep.
- High tide occured at 2 pm, when the bay was 15 m deep.

a Let t be the number of hours passed since low tide was first measured. Complete the following table of values for the tide level:

Time (t in hours)	0	3	6	9	12
Tide level (d in meters)					

b Find the amplitude of the tide level.

c Find the period of the tide level.

d Sketch a graph of the tide level, d, over the first 12 hours.

e State whether the following equations can be used to model the tide level, d, at t hours after 8 am, where the parameters A, n, and C are positive:

i $d = -A\cos nt + C$

ii $d = A\cos nt + C$

iii $d = A\sin nt + C$

iv $d = -A\sin nt + C$

f Find an equation for d.

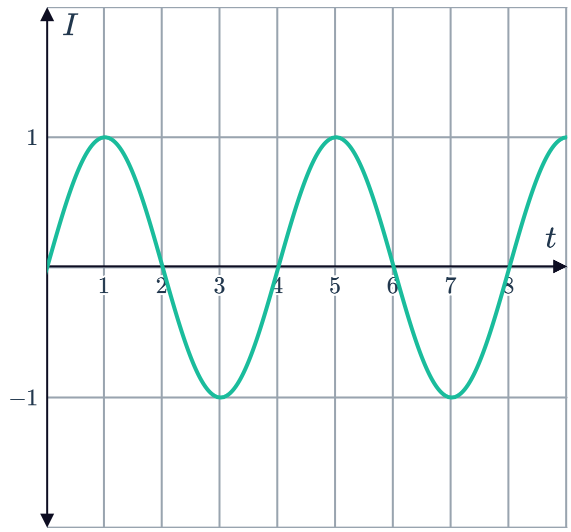
g The ship needs a depth of at least 11.7 m to be able to sail into the bay safely. Find the number of hours Brad has after high tide before he can no longer safely sail into the bay. Round your answer to two decimal places.

36 Sounds around us create pressure waves. O[redacted]rs interpret the amplitude and frequency of these waves to make sense of the sounds. A speaker is set to create a single tone, and the graph below shows how the pressure intensity (I) of the tone, relative to atmospheric pressure, changes over t seconds.

a) State whether a sine or cosine function be more suited for modeling this graph. Explain your answer.

b) Find the equation of the function for the intensity I after t seconds.

c) As a brain exercise, a music student must tap their foot every time they hear the most intense part of the sound. How many times will the student tap their foot in 14 minutes?



37 The change in voltage of overhead power lines over time can be modeled by a periodic function that oscillates between -300 and 300 kiloVolts with a frequency of 40 cycles per second.

a) Assuming that at $t = 0$ seconds, the voltage is 300 kV, state whether the following equations can be used to model voltage over time:

i) $V = A \cos nt$ ii) $V = A \sin nt$

iii) $V = A \sin t$ iv) $V = A \cos t$

b) Find an equation for the voltage V as a function of time t.

c) Sketch a graph of the voltage over time for $0 \leq t \leq \dfrac{1}{16}$.

d) Describe the effect on the graph of voltage over time if the frequency was increased.

38 12 seconds into an opera song, there is a section in which the singer must alternate between high and low pitch notes for 6 seconds. During this part of the song, the volume v of the sound is given by the function $v\left(t\right) = 6 \sin\left(\dfrac{\pi}{2} t\right) + 24$.

a) Sketch a graph the volume function for the 6-second section.

b) Find the maximum volume the singer's voice reaches during the section.

c) What does t represent in this function?

d) Find the volume of the singer's voice 2 seconds into the section.